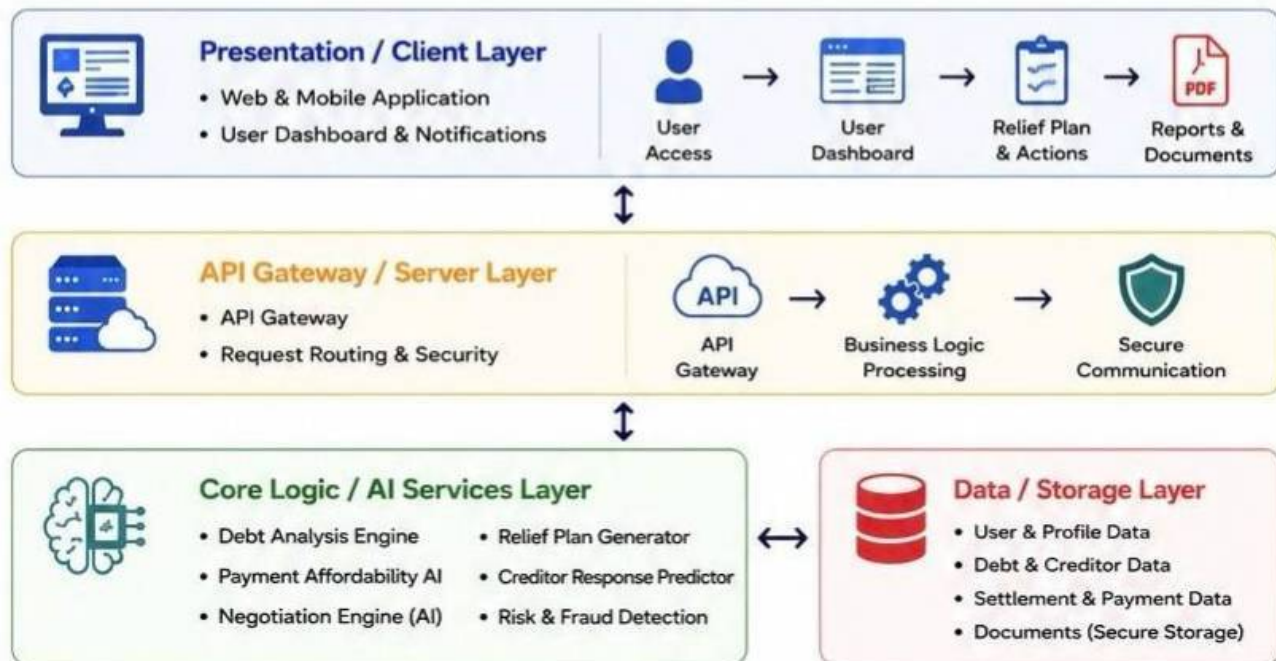


Solution Architecture – AI Debt Relief

 Date	10 July 2026
 Team Code	XXXXXXX
 Project Name	AI Debt Relief Platform
 Maximum Marks	5 Marks

Solution Architecture Diagram

The AI Debt Relief Platform automates debt assessment, negotiation and settlement to help users become debt-free faster. It uses AI and data analytics to analyze debts, create personalized relief plans, negotiate with creditors and track settlements securely.



Data Flow Overview



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides user interface for onboarding, debt overview, relief plans, payments and communication.	HTML5, CSS3, React.js, Tailwind, Firebase
API Gateway / Server	Handles API requests, authentication, rate limiting, routing and load balancing.	Python, FastAPI, Nginx, Uvicorn
Core Logic / AI Services	Performs debt analysis, affordability calculation, AI negotiations and relief plan generation.	Python, TensorFlow, Scikit-learn, LangChain, Pandas, NumPy
Data / Storage Layer	Stores user data, debt information, creditor data, transactions, documents and logs securely.	PostgreSQL, Redis, AWS S3, MongoDB (optional)
External Services	Integrations with credit bureaus, payment gateways, email/SMS and document verification.	Experian API, Plaid, Twilio, SendGrid, Stripe

Component Description & Technologies

Component Name	Description / Role in Architecture	Technologies Used
Authentication Module	Manages user registration, login, KYC verification and role-based access control.	JWT, OAuth 2.0, Firebase Auth, Auth0
AI Analysis & Scoring Engine	Analyzes debt profile, payment behavior and affordability to generate relief recommendations.	Python, Scikit-learn, XGBoost, Pandas
Negotiation Engine	AI-driven engine to negotiate with creditors and suggest best settlement offers.	Python, LangChain, NLP, GPT-4 API (optional)
Notifications Service	Sends email, SMS and in-app notifications to keep users updated.	Twilio, SendGrid, Firebase Cloud Messaging
Reporting & Documents	Generates settlement letters, reports, payment schedules and downloads.	PDFKit, WeasyPrint, Docx, AWS S3
Payment & Settlement Tracking	Tracks payments, settlements and updates debt status in real-time.	Stripe, Plaid, PostgreSQL, Redis

Key Benefits



Smarter Relief

AI analyzes debt and suggests personalized relief strategies.



Time Saving

Automated negotiations and workflows reduce manual effort.



Higher Success

Better settlement outcomes through AI-driven insights.



Secure & Trusted

End-to-end data encryption and secure document handling.